Disciplined Heterogeneity in the Rainfall–Mortality Response

Allowing many margins, then selecting among them

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The Problem

Two mechanisms, one model

We want a specification that lets **both** narratives exist and be estimated on the full sample:

- **Extreme rainfall** damages — flooding, trauma, waterborne disease. Operates at a *daily* scale, days-to-weeks lags, concentrated where drainage and density make flooding lethal.
- **Drought** damages — water scarcity, lost agricultural income, malnutrition. Operates on *accumulated deficits* over months, with lags of 6–18 months, concentrated where water infrastructure is poor and livelihoods are rain-dependent.

A single smooth f(daily rainfall) **cannot represent both**, because they live at different temporal scales. Worse, several channels are non-monotone in opposite directions (diarrhoeal disease is U-shaped in rainfall; vector-borne disease is hump-shaped), so an all-cause polynomial can average genuine effects toward zero.

And a search problem we have already been burned by

Testing many candidate margins one at a time does not work. From the PHL analysis:

- Five urban-share bins $\times$ three ages gave **one** significant cell out of fifteen — against ≈ 0.75 expected by chance.
- **Which** cell was significant *moved when the bins were redrawn*: quintiles flagged (88.8–100]; a finer grid flagged (90–95] while (95–100] went quiet; the *least* urban bin cleared for 65+.
- Formal tests of the implied threshold — top-vs-bottom, top-vs-pooled-rest, and a scan over cutoffs from 60% to 95% — were significant **nowhere** (smallest $p = 0.11$ across 18 tests).

We need a procedure that can consider many margins *without* inheriting the multiple-comparison problem of having looked at all of them.

Specification

A model that nests both mechanisms

$$D_{ait} = \exp\Big(\underbrace{f_a(\mathbf{E}_{it})}_{\text{daily extremes}} + \underbrace{g_a(\mathbf{S}_{it})}_{\text{accumulated deficit}} + \sum_j \left[f_a^j(\mathbf{E}_{it}) V_{ij}^{\text{flood}} + g_a^j(\mathbf{S}_{it}) V_{ij}^{\text{dry}} \right] + \mu + \nu \Big)$$

- **E**: daily-scale exposure — days above local p95/p99, excess rain above threshold, Rx1day / Rx5day.
- **S**: accumulated deficit — SPI/SPEI at 6 and 12 months, log-deviation over rolling windows, consecutive dry days.
- **The two exposures carry different interaction sets.** Forcing one vulnerability dimension to serve both is precisely what stops the current specification from representing both stories.

Additive separability in scale is the key restriction. A cross term (drought raising flood vulnerability) is allowed but expected to be second-order.

Selection should do less work than it sounds like

The strongest discipline is **over-identification**, not fit. Each mechanism makes four independent, falsifiable predictions:

	Flood mechanism	Drought mechanism
Functional shape	wet tail, daily intensity	accumulated deficit
Timescale	days–weeks	6–18 months
Cause of death	drowning, injury, leptospirosis	intestinal, malnutrition
Vulnerability	drainage, density, poverty	water infra., aridity, poverty

A model that *fits* best is weak evidence. A mechanism that correctly predicts **shape, lag, cause and heterogeneity simultaneously** is strong evidence, because there are far more testable implications than free parameters. Let theory restrict the space; use ML to choose within it.

Method

Residualise, then LASSO

1. **Build wide.** Every exposure term $\times$ age $\times$ every candidate vulnerability variable, in *levels and squares* so non-monotone heterogeneity is expressible.
2. **Partial out the fixed effects** by Frisch–Waugh (`fixest::demean`). The LASSO never sees the >18,000 fixed effects; they are absorbed exactly.
3. **LASSO the interactions, never the core.** `penalty.factor = 0` on f_a and g_a: the objects of interest are always in and never shrunk. Only the heterogeneity space is penalised.
4. **Choose λ by cluster-blocked CV** — folds are whole provinces.
5. **Estimate** the selected model by `fepois`, and assess stability by bootstrap.

Three details that decide whether this works

- **Cluster-blocked folds are not optional.** i.i.d. CV lets the same municipality inform training and validation; under spatial and serial correlation it systematically *underpenalises* and will select noise. Folds must be whole provinces (or the theory-driven plug-in λ of Belloni et al.).
- **Correlated terms are selected arbitrarily.** Post-FE, the linear and quadratic precipitation terms correlate at 0.79. LASSO keeps one essentially at random and swaps between folds. Use *group* LASSO so a polynomial block enters or exits together, or keep functional form out of the selection entirely.
- **Naive post-LASSO inference is invalid.** Reporting selected coefficients with ordinary standard errors gives shrunk estimates and p-values that ignore the search. Valid options: post-double selection (Belloni–Chernozhukov–Hansen), cross-fitted double ML, or — as here — report *selection frequencies* rather than p-values.

Discovery and confirmation on different data

Regularisation controls overfitting *within* a sample. It does not protect against the risk that a margin is specific to one country.

Split	Share	Use
Develop	$\sim$40%	find candidate margins freely; nothing reportable
Tune	$\sim$20%	choose λ, bin edges, lag lengths
Confirm	$\sim$40%	estimate once; **these** are the reported numbers

Split by geography, not at random. Random splitting leaks: the same municipality appears in both halves and spatial correlation carries information across. With a multi-country panel, develop on one country and confirm on the others — out-of-sample in the sense that matters for the claim we want to make.

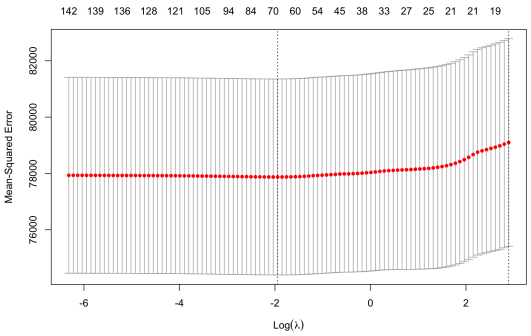
Prototype: Philippines

The prototype design

Run on the PHL panel to stress-test the machinery before the multi-country data arrives (`05_lasso_heterogeneity.R`).

- **82,593** municipality × year × age observations.
- **7 vulnerability variables**: urban share, coastal share, log income, long-run precipitation, long-run temperature, log population, settlement class — each in levels and squares.
- **144 design columns**: 18 core exposure terms (unpenalised) + **126 penalised interactions**.
- Fixed effects absorbed by Frisch–Waugh: **13,041** municipality × age × era and **5,187** province × year × age.
- λ chosen by 10-fold CV with **provinces as folds**.

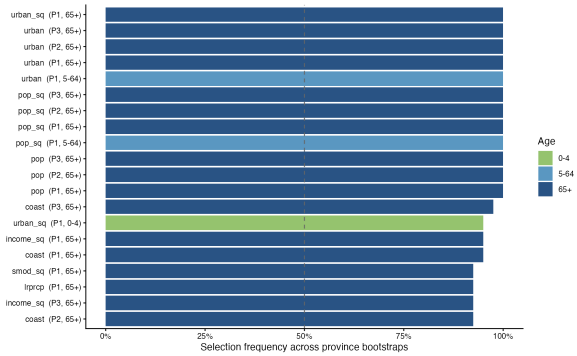
Result: the conservative rule selects nothing



- $\lambda_{1se} = 18.1$ selects **0 of 126** interactions.
- $\lambda_{min} = 0.143$ selects **49**.
- The two differ by **two orders of magnitude** — the CV curve is very flat, which is what a weak heterogeneity signal looks like.

Under the standard conservative rule, **no heterogeneity in the PHL rainfall response survives province-blocked cross-validation**. That is consistent with everything the bin-by-bin analysis found, and it is the honest headline.

Which margins survive at the permissive penalty



Selection frequency over **40 province-block bootstraps** at $\lambda_{\min}$. **22 of 114** terms are selected in $\geq 90\%$ of resamples.

Ordering by dimension: **population** and **urban share** first (both at 100%), then coastal, income, long-run precipitation.

Read with care. These frequencies are conditional on $\lambda_{\min}$, a deliberately permissive penalty chosen because the 1-s.e. rule kept nothing. High stability here means "robustly selected *given* a generous penalty", not "significant".

What the prototype actually teaches

- **The machinery works** and scales: 126 interactions and 18,000 fixed effects handled in one pass, minutes of compute.
- **Population size emerges as a margin we had not tested.** It was in no bin-by-bin analysis, and it is selected in every bootstrap. It is also correlated with urban share — so part of what we read as an "urban" effect may be a *population-size* effect, which has different mechanisms (health-system capacity, registration quality, small-count noise).
- **The PHL null is now much better established.** Not "we tried some bins and found little", but "a penalised search over 126 candidate margins, disciplined by spatially blocked CV, retains nothing under the conservative rule".
- That is a far stronger prior to carry into the multi-country panel, where the same procedure has real variation to work with.

Caveats and Next Steps

What is approximate here

- **FWL is exact for linear models; our estimator is Poisson.** Selection currently happens in the linear analogue and estimation in the Poisson model. Defensible — selection need not use the estimation likelihood — but not exact. The clean fix is demeaning inside the IRLS loop, or a Poisson LASSO with offsets.
- **Bootstrap stability is not inference.** It answers "how reliably is this term selected", not "is this coefficient non-zero". Valid inference needs post-double-selection or cross-fitting, which is the next build.
- **Correlated polynomial terms should be grouped.** Currently selected individually; group LASSO is the correct object.
- **The heterogeneity space is still small.** Seven variables, one exposure scale. The design only becomes interesting once daily extremes and accumulated deficits are both present.

Next steps

1. **Daily and monthly climate** from the cluster — without both exposure scales the two-mechanism model cannot be written down.
2. **Cause of death**, which is what separates cancelling mechanisms and gives each its own functional form.
3. **Water and sanitation coverage**, the vulnerability margin the drought literature says governs the deficit channel.
4. **Group LASSO plus cross-fitting**, for valid inference on the selected margins.
5. **Add Brazil and Mexico**, and adopt the develop/confirm split across countries — develop the specification in one, confirm in the others.

Step 5 is what turns this from a selection procedure into evidence: a margin that is discovered in one country and confirmed in another is a different kind of claim from one that survives a penalty in the sample where it was found.